

Part2 51Voxel

Friday, August 22, 2025 9:19 AM

RESOURCES for today's workshop....

GUI Dataset Creator

https://github.com/harpreetsahota204/gui_dataset_creator

COCO4GUI

https://github.com/harpreetsahota204/coco4gui_fiftyone

Code and Notebooks

https://github.com/harpreetsahota204/visual_agents_workshop

FiftyOne Plugins

https://github.com/harpreetsahota204/synthetic_gui_samples_plugins

Slides

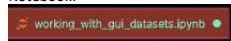
<https://docs.google.com/presentation/d/1TyaiSnMVQGneDbarzCFb8nTAudg2aCgUikpRGq8ibv4/edit?usp=sharing>

Teja - you can find the playbacks from this series here: <https://www.youtube.com/playlist?list=PLuREAXoPgTQmbAU09VDO5pTi4EikeGeD>

myNotes

Annotating and Organizing Datasets walk-thru

Notebook:



Created views into Dataset, can now view in the App

```
For more details and examples, see the Filtering Cheat Sheet and Views Cheat Sheet.
```

```
1 # Create a view with missing labels
2 %!( to chat, %K to generate
3 no_bb_labels = gui_dataset.filter_labels("detections", F("label") == "")
4
5 gui_dataset.save_view("no_bb_labels", no_bb_labels)
```

```
[ ]
```

You can also do the same with keypoints as well:

Compute all BoundingBoxes (BBs)

Analyzing Bounding Box Areas

To compute the area of bounding boxes in FiftyOne, you can use a [ViewExpression](#). Bounding boxes are stored in the format [top-left-x, top-left-y, width, height], where width and height are relative to the image size (values between 0 and 1).

This computes the area as width × height in relative coordinates (as a fraction of the image).

You can use these expressions in view stages like `filter_labels` to filter detections by area.

```
1 from fiftyone import ViewField as F
2
3 rel_bbox_area = F("bounding_box")[2] * F("bounding_box")[3]
4
5 im_width, im_height = F("$metadata.width"), F("$metadata.height")
6 %!( to chat, %K to generate
7 gui_dataset.set_field("detections.relative_bbox_area", rel_bbox_area).save()
```

Python

Build a similarity index:

Build powerful similarity indexes for:

- Finding visually similar GUI screens
- Identifying related interaction patterns
- Retrieving nearest neighbors for any sample

FiftyOne supports multiple backends including:

- Sklearn (default, in-memory)
- Vector databases (Qdrant, Pinecone, Redis, Milvus, etc.)
- Document stores (MongoDB, Elasticsearch)

```
1 results = f1.compute_similarity(
2     gui_dataset,
3     backend="sklearn", # Fast sklearn backend
4     brain_key="clip_sim",
5     embeddings="clip_embeddings"
6 )
```

```
[ ]
```

Compute a representative score:

Computing Representativeness

Create diverse subsets of your GUI dataset by:

- Selecting samples that best represent the entire distribution
- Ensuring coverage of all interaction types and patterns
- Avoiding redundancy in your selected examples

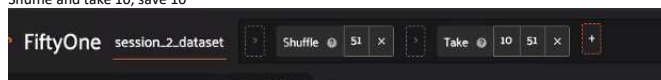
```
1 results = fob.compute_representativeness(  
2     gui_dataset,  
3     similarity_index="clip_sim"  
4 )
```

Semantic patches

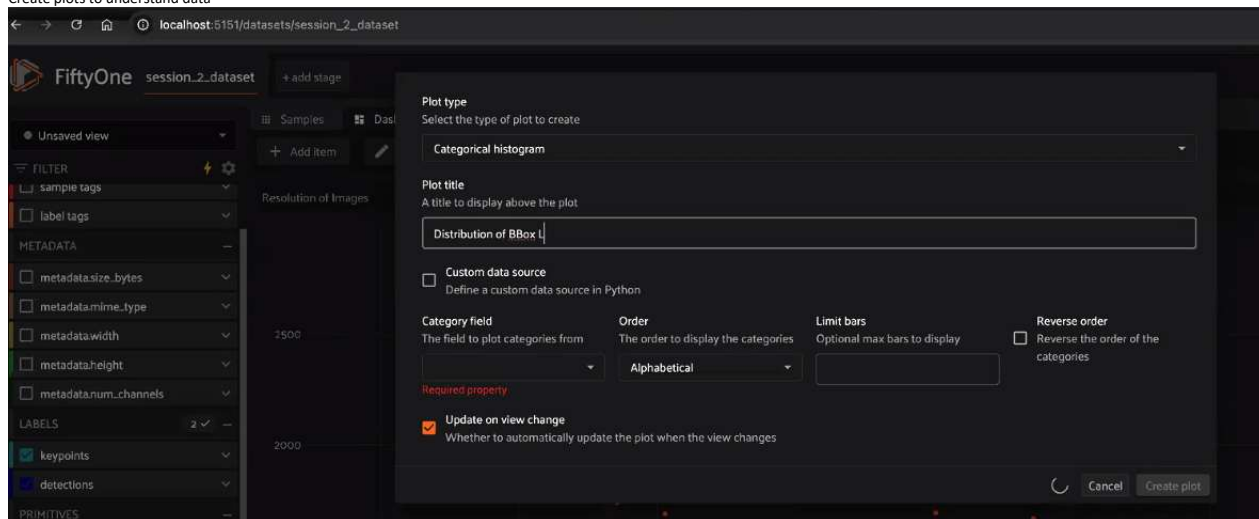
```
1 from sentence_transformers import SentenceTransformer  
2 from tqdm import tqdm  
3  
4 # Initialize the embedding model  
5 model = SentenceTransformer("nomic-ai/nomic-embed-text-v1.5", trust_remote_code=True, device="mps")  
6  
7 # Iterate through patches and compute embeddings  
8 for patch in tqdm(box_patches, desc="Computing embeddings"):  
9     # Construct prompt from the three fields  
10    task_desc = patch.detections.task_description if patch.detections.task_description else ""  
11    action_type = patch.detections.action_type if patch.detections.action_type else ""  
12    element_info = patch.detections.element_info if patch.detections.element_info else ""  
13  
14    # Combine into prompt  
15    prompt = f"clustering: Task: {task_desc}; Action: {action_type}; Element: {element_info}"  
16  
17    # Compute embedding  
18    embedding = model.encode(prompt)  
19  
20    # Add as new field  
21    patch['text_embedding'] = embedding.tolist()  
22    patch.save()  
23  
24 print("Done! Embeddings added to all patches.")
```

Command: fiftyone app launhc
Session2 DS

Shuffle and take 10, save 10



Create plots to understand data



Add to the side, now have a dashboard panel



Resolution scaling

- Simulate different screen sizes (laptop, desktop, 4K)
- Test model robustness across devices
- No coordinate transformation needed
- Coverage across various resolution ranges

-Color inversion
-Colorblindness simulation
-grayscale conversion
-Task description augmentation with LLMs

- Handles different phrasings of same task
- Maintains semantic meaning
- Prepares for varied user instructions



At 10:11 am

Homework

Next week: train a model, fine-tune a model to use 51 or using 51?

Use the model, train a model to use 51? \$actodo if comeback to this

Q&A:

Andre Milota

Anyone want to work on building some AI and agentic programs. I have a meetup where we try to pair

program some things like:
Agents, Lang Graph, Multimodal UI, vision, neural networks
I have lots of projects in mind and am looking for people to work with who are also trying to master this technology.
<https://www.linkedin.com/in/andremilota>

Q: cheat sheets?

Visual Agents Workshop Part 1

harpreetsahota204/gul_data

harpreetsahota204/coco4gu

FiftyOne

FiftyOne User Guide — FiftyO

docs.voxel51.com/user_guide/

VOXEL51

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Exporting datasets

Images that do not contain dogs:

```
ds.match(~F("ground_truth.detections.label").contains("dog"))
```

Dates and Times

After a specific date:

```
ds.match(F("date") > query_date)
```

Within 30 minutes of a specific date:

```
ds.match(abs(F("date") - query_date) < query_delta)
```

Geospatial

Ask a follow-up

Deep thinking

Powered by kapa.ai

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Let's connect



LinkedIn



GitHub



X